NeuroFuzzy: A Soft Computing Pipeline for Brain Tumor MRI Diagnosis

Project report: a five-stage soft computing pipeline for brain MRI classification

Methods: Fuzzy C-Means, GLCM, Binary PSO, neuro-fuzzy network (PyTorch), Mamdani fuzzy risk scoring, Kohonen SOM, FGSM adversarial test

1. Overview

The project builds a diagnostic pipeline for brain MRI scans that separates healthy tissue from tumor. Each stage is a soft computing method, so the reasoning can be inspected: fuzzy clustering for segmentation, texture features, a swarm search to remove redundant features, a neuro-fuzzy classifier, a fuzzy rule base that turns the output into a risk score, and a self-organising map that checks whether the features form natural groups.

This report describes what the notebook does and what it produced. The results in Section 4 are from the saved run in *Code.ipynb*. They are not clinical results, and Section 5 lists the reasons.

2. Data

The dataset folder holds four MRI classes (glioma, meningioma, notumor, pituitary) with 1,400 training and 400 test JPEG images per class. The notebook labels *notumor* as healthy (0) and the other three as tumor (1). It uses a subset: 150 training images and 50 test images per class, giving 600 training and 200 test images. The test set has 50 healthy and 150 tumor images.

The dataset's origin and licence should be checked before the project is published. The file naming matches a public Brain Tumor MRI dataset with these four classes, but this report does not confirm the source.

3. Method

Stage	What the notebook does
1. Segmentation	Resizes each MRI to 128x128 and runs Fuzzy C-Means (2 clusters, m=2). The brighter cluster is taken as the tumor and thresholded at membership 0.5.
2. Features	GLCM texture (contrast, correlation, energy, homogeneity at 4 angles) plus mean intensity, standard deviation and area of the segmented region: 19 features.
3. Feature selection	Binary PSO (mealpy OriginalPSO, 30 epochs, 20 particles). Fitness is 3-NN accuracy minus a small penalty per selected feature, scored on a validation split taken from the training data (the test set is not used). The result keeps 8 of the 19 features: contrast_0, contrast_2, correlation_2, energy_0, energy_2, homogeneity_2, mean_intensity, std_intensity.
4. Classifier	Neuro-fuzzy network with 10 Gaussian rules, a product T-norm, sigmoid output, trained for 500 epochs with Adam (lr 0.01) and binary cross-entropy.
5. Risk score	Mamdani system: the network probability and a 0 to 10 severity from the tumor area feed five rules, giving a risk score from 0 to 100%.
6. SOM	15x15 Kohonen map over the 8 selected features, trained on training data for 1,500 iterations. Test labels are overlaid on the map for display only.
7. Adversarial test	FGSM attack on the test features with epsilon 0.1, measuring the accuracy drop against a 15-point cutoff.

4. Results (latest run, after the validation fix)

Measure	Result
Test accuracy, classifier on 8 selected features	88.50% (200 images)
Healthy class: precision / recall / F1 (support 50)	0.79 / 0.74 / 0.76
Tumor class: precision / recall / F1 (support 150)	0.92 / 0.93 / 0.92
Accuracy under FGSM attack, epsilon 0.1	81.50%
Accuracy drop under attack	7.00 points. The notebook reports SECURE, using a 15-point cutoff.
Mamdani risk for one true-positive test case	67.88% (network probability 0.745, severity 5.95)
Feature reduction	19 to 8 features

The earlier run, before the validation split, reported 88.00% accuracy and a 17.50-point drop (VULNERABLE). That run used the test set to choose features, so its numbers are not reliable and are not used here.

The SOM output is a visual map. The notebook reports that healthy and tumor points form separate regions, but it gives no number for this. The Mamdani result is one example case, not an evaluation.

5. Limitations

• **Single split and single run.** Results come from one train/test split and one run. There is no cross-validation or external test set.

• **Subset of the data.** Only 150 training and 50 test images per class are used.

• **Tumor segmentation is a heuristic.** The brightest FCM cluster is assumed to be the tumor, for every image, including healthy scans. The area feature is therefore not a true tumor measure for healthy images.

• **Binary task.** The three tumor types are merged, so the model does not identify the type of tumor.

• **Coarse risk score.** The example case gives 67.88% for network probabilities 0.958 and 0.745, and 50% for 0.2 and 0.5. The rule base does not respond to changes in the probability over these ranges, so the Mamdani stage needs its membership functions reviewed.

• **Adversarial verdict depends on the cutoff.** The notebook marks the model SECURE because the drop is under 15 points. Accuracy still falls by 7 points under a small attack, which should be stated with the result.

• **Not for clinical use.** No clinician review, regulatory validation or prospective study has been done.

6. Repository contents and setup

The dataset is not stored in the repository because of its size. Download it from its original source and place it in *dataset/*, keeping the Training and Testing folders and the four class folders in each.

```
python -m venv venv
source venv/bin/activate
pip install -r requirements.txt
jupyter notebook Code.ipynb
```

7. Conclusion and next steps

The pipeline runs end to end. After the validation fix, the classifier reaches 88.50% on the held-out test set with 8 features. The main next steps are: (1) run cross-validation on the full dataset; (2) review the Mamdani membership functions, since the risk score barely changes across a wide range of probabilities; (3) check the segmentation heuristic on healthy images; (4) justify the 15-point adversarial cutoff or replace it with a stated threshold.

With these checks done, the results can be reported with confidence intervals and compared with the base paper's figures.